Helin Wang

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Basic Information

Home Page: https://wanghelin1997.github.io/helinwang/
Google Scholar Page: https://scholar.google.com/citations?user=I_V0zBMAAAAJ
Github Page: https://github.com/WangHelin1997

Research Interest

My research interest majorly lies in **AI for speech and audio signal processing**.

Education

Johns Hopkins University, Baltimore, USA — 2022 – Present
Ph.D student in Whiting School of Engineering (WSE)
Supervisor: Najim Dehak

Peking University, Beijing, China — 2019 – 2022
Master student in School of Electronic and Computer Engineering (ECE)
Supervisor: Yuexian Zou

Tsinghua University, Beijing, China — 2015 – 2019
B.S. in Department of Automation (DA)

Experience

Johns Hopkins University, CLSP, Baltimore, USA — August 2022 – Present
Research Assistant Supervisor: Najim Dehak, Laureano Moro-Velázquez and Jesús Villalba

Tencent AI Lab, Speech Group, Bellevue, USA — May 2024 - August 2024
Intern Supervisor: Meng Yu and Dong Yu

Microsoft STCA, NLP Group, Beijing, China — February 2022 - May 2022
Intern Supervisor: Linjun Shou and Ming Gong

Tencent AI Lab, Speech Group, Shenzhen, China — May 2020 - November 2021
Intern Supervisor: Bo Wu and Chao Weng

Peking University, ADSP LAB, Shenzhen, China — August 2019 - July 2022
Research Assistant Supervisor: Yuexian Zou

Ubtech Robotics Inc, Speech Group, Shenzhen, China — July 2019 - September 2019
Intern Supervisor: Dongyan Huang

University of California Berkeley, Berkeley, USA — July 2018 - September 2018
Summer Researcher Supervisor: Masayoshi Tomizuka

Selected Publications

1. **Helin Wang***, Jiarui Hai*, Yen-Ju Lu, Karan Thakkar, Mounya Elhilali, Najim Dehak: SoloAudio: Target Sound Extraction with Language-oriented Audio Diffusion Transformer. Submitted to ICASSP 2025.
2. **Helin Wang**, Meng Yu, Jiarui Hai, Chen Chen, Yuchen Hu, Rilin Chen, Najim Dehak, Dong Yu: SSR-Speech: Towards Stable, Safe and Robust Zero-shot Text-based Speech Editing and Synthesis. Submitted to ICASSP 2025.
3. **Helin Wang**, Jesus Villalba, Laureano Moro-Velazquez, Jiarui Hai, Thomas Thebaud, Najim Dehak: Noise-robust Speech Separation with Fast Generative Correction. INTRESPEECH 2024 (**Oral**). ⋆ Nominated for the **Best Paper Award** and the **Best Student Paper Award** (5 out of 1,030 accepted papers).
4. Jiarui Hai*, **Helin Wang***, Dongchao Yang, Karan Thakkar, Najim Dehak, Mounya Elhilali: DPM-TSE: A Diffusion Probabilistic Model for Target Sound Extraction. ICASSP 2024: 1196-1200.
5. **Helin Wang**, Venkatesh Ravichandran, Milind Rao, Becky Lammers, Becky Lammers, Myra Sydnor, Nicholas Maragakis, Ankur A. Butala, Jayne Zhang, Lora Clawson, Victoria Chovaz, Laureano Moro-Velazquez: Improving fairness for spoken language understanding in atypical speech with Text-to-Speech. NeurIPS 2023 Workshop on Synthetic Data Generation with Generative AI (**Oral**).
6. **Helin Wang**, Thomas Thebaud, Jesus Villalba, Myra Sydnor, Becky Lammers, Najim Dehak, Laureano Moro-Velazquez: DuTa-VC: A Duration-aware Typical-to-atypical Voice Conversion Approach with Diffusion Probabilistic Model. INTERSPEECH 2023: 1548-1552.
7. Dading Chong*, **Helin Wang***, Peilin Zhou, Qingcheng Zeng: Masked Spectrogram Prediction For Self-Supervised Audio Pre-Training. ICASSP 2023: 1-5.
8. Dongchao Yang*, **Helin Wang***, Zhongjie Ye, Yuexian Zou, Wenwu Wang: RaDur: A Reference-aware and Duration-robust Network for Target Sound Detection. INTERSPEECH 2022: 1511-1515.
9. **Helin Wang**, Dongchao Yang, Chao Weng, Jianwei Yu, Yuexian Zou: Improving Target Sound Extraction with Timestamp Information. INTERSPEECH 2022: 1526-1530.
10. **Helin Wang**, Yuexian Zou, Wenwu Wang: A Global-Local Attention Framework for Weakly Labelled Audio Tagging. ICASSP 2021: 351-355.
11. **Helin Wang**, Yuexian Zou, Wenwu Wang: SpecAugment++: A Hidden Space Data Augmentation Method for Acoustic Scene Classification. INTERSPEECH 2021: 551-555.
12. **Helin Wang**, Bo Wu, Lianwu Chen, Meng Yu, Jianwei Yu, Yong Xu, Shi-Xiong Zhang, Chao Weng, Dan Su, Dong Yu: TeCANet: Temporal-Contextual Attention Network for Environment-Aware Speech Dereverberation. INTERSPEECH 2021: 1109-1113.
13. **Helin Wang**, Yuexian Zou, Dading Chong, Wenwu Wang: Modeling Label Dependencies for Audio Tagging With Graph Convolutional Network. IEEE Signal Process. Lett. 27: 1560-1564 (2020).
14. **Helin Wang**, Yuexian Zou, Dading Chong, Wenwu Wang: Environmental Sound Classification with Parallel Temporal-Spectral Attention. INTERSPEECH 2020: 821-825.

Services (Reviewer)

- IEEE/ACM Transactions on Audio, Speech, and Language Processing
- IEEE Signal Processing Letters
- Neurocomputing
- INTERSPEECH
- ICASSP
- NeurIPS

Teaching

2024/02 - 2024/05, Teaching Assistant, Johns Hopkins University, Baltimore, USA:
EN.520.123: Computational Modeling for Electrical and Computer Engineering in Spring 2024 in the Department of Electrical and Computer Engineering

Honors and Awards

1st Team Ranking of DCASE Challenge Task 5 (Judges' award) 2021

4^{th} *Team Ranking* of DCASE Challenge Task 6 (Judges' award)	2021
3^{rd} *Team Ranking* of DCASE Challenge Task 6	2020
Outstanding Graduate Student Award of Peking University	2022
Outstanding Graduate Thesis Award of Peking University	2022
Award for Scientific Research of Peking University	2020-2021
School Prize of Peking University	2019-2020
Merit Student of Peking University	2019-2020

Skills

- Programming Languages: Python, Matlab
- Platform: Linux
- English: IELTS Test Overall 7.0
- Others: Pytorch, Tensorflow